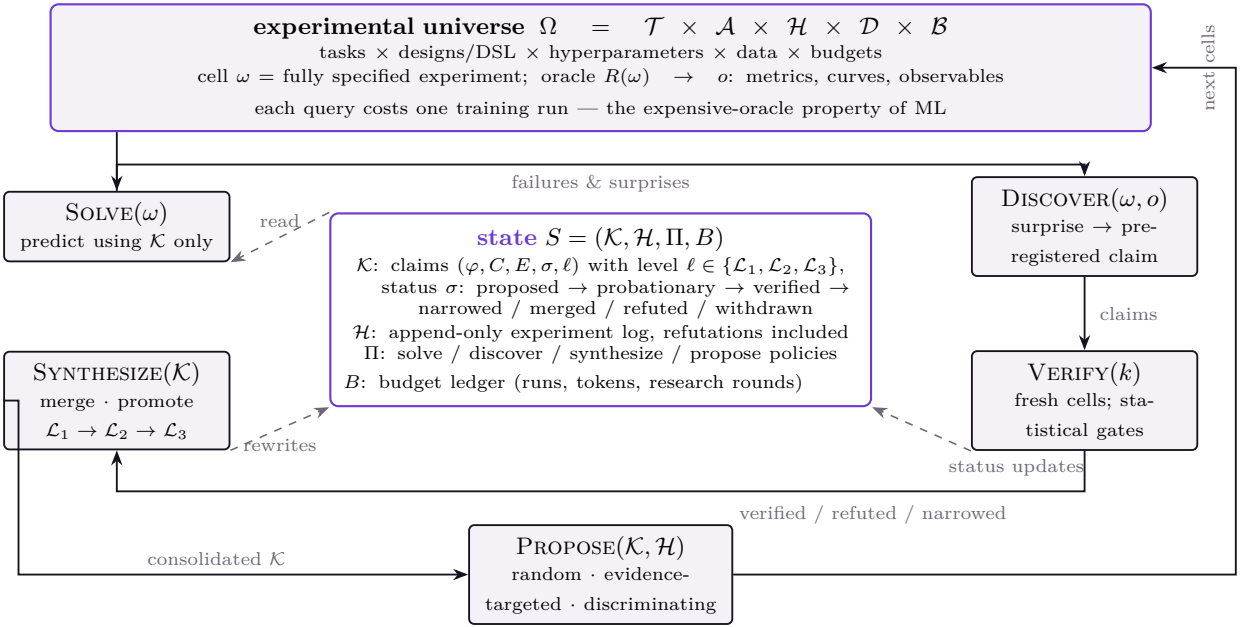

discipline invariants (each checkable from the state): **(i) freshness** — verify only on cells created after registration; **(ii) pre-registration** — predictions committed before runs; **(iii) sealed sets** — held-out viewed once, then development; **(iv) evidence immortality** — refutations and failed predictions retained